

Linear Programming Report

Sensitivity Analysis & Duality

Infeasible Primal / Unbounded Dual

Same constraints with $x \geq 0$. The primal is infeasible and the dual ($\max y_1 + y_2$, $y_1 - y_2 \leq 0$ and ≤ 1) is unbounded along $y_1 = y_2$.

Source: Lecture slide 56

Problem type	MIN, 2 variables, 2 constraints
Status	INFEASIBLE
Optimal Objective	–
Dual status	UNBOUNDED
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1. Model formulation

```

MIN   z = x1
subject to
    x1 + x2 >= 1      (Con1)
    -x1 - x2 >= 1     (Con2)
    x1 >=0, x2 >=0
  
```

Data table

Constraint	x1	x2	Sense	RHS
Con1	1	1	>=	1
Con2	-1	-1	>=	1
Objective c	1	0	min	–

Definitions

- **x1**: variable 1
- **x2**: variable 2

2. Result

The problem is **infeasible**. Its dual is **unbounded**. An infeasible primal has an infeasible or unbounded dual.

Dual formulation

```

MAX   z = y_Con1 + y_Con2
subject to
    y_Con1 - y_Con2 <= 1      (x1)
    y_Con1 - y_Con2 <= 0     (x2)
    y_Con1 >=0, y_Con2 >=0
  
```

